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Cricket Fielding Analysis - Data Collection & Performance Score Calculator

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Single-file version: everything (docs, sample data, calculation, CSV/chart export) lives in this one script so it's easy to drop into a repo.

WHAT IT DOES

1. Stores ball-by-ball fielding events in the exact "Dataset Features" format from the assignment (Match No, Innings, Team, Player Name, Ballcount, Position, Short Description, Pick, Throw, Runs, Overcount, Venue), plus a Direct Hit flag.
2. Aggregates each player's raw counts: Clean Picks (CP), Good Throws (GT), Catches (C), Dropped Catches (DC), Stumpings (ST), Run Outs (RO), Missed Run Outs (MRO), Direct Hits (DH), and Runs Saved (RS).
3. Applies the Performance Score formula:

$$\begin{aligned} \text{PS} = & (\text{CP} \times \text{WCP}) + (\text{GT} \times \text{WGT}) + (\text{C} \times \text{WC}) + (\text{DC} \times \text{WDC}) + (\text{ST} \times \text{WST}) \\ & + (\text{RO} \times \text{WRO}) + (\text{MRO} \times \text{WMRO}) + (\text{DH} \times \text{WDH}) + \text{RS} \end{aligned}$$

4. Prints the raw data, the performance metrics matrix, and a ranking to the console, and saves:
 - raw_fielding_data.csv (ball-by-ball log)
 - performance_summary.csv (per-player metrics + PS)
 - performance_scores.png (bar chart comparing the three players)

WEIGHTS

The assignment sheet doesn't print numeric weight values, so sensible defaults are defined below. Edit the WEIGHTS dict to match whatever values your instructor supplies.

WCP = 1.0 Clean Pick
WGT = 1.0 Good Throw

WC = 2.0 Catch
WDC = -2.0 Dropped Catch
WST = 2.0 Stumping
WRO = 3.0 Run Out
WMRO = -1.0 Missed Run Out
WDH = 1.5 Direct Hit

SAMPLE DATA

RAW_EVENTS below is SYNTHETIC sample data (placeholder player names, a fictional match) that exercises every category in the formula. Replace it with your own observed match data before submitting the assignment -- the rest of the pipeline (aggregation, scoring, CSV/chart export) works unchanged.

REQUIREMENTS

```
pip install matplotlib
```

RUN

```
python fielding_analysis.py
```

SAMPLE OUTPUT

RANKING

1. Liam O'Brien: 15.0
2. Rahul Sharma: 8.5
3. Steve Walker: -2.0

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```
import csv
import os
from collections import defaultdict
```

```
import matplotlib
```

```
matplotlib.use("Agg") # headless-safe backend for saving PNGs
import matplotlib.pyplot as plt
```

```
# -----
# 1. Weights used in the Performance Score formula (edit as needed)
# -----
```

```
WEIGHTS = {
    "WCP": 1.0, # Clean Pick
    "WGT": 1.0, # Good Throw
    "WC": 2.0, # Catch
    "WDC": -2.0, # Dropped Catch
    "WST": 2.0, # Stumping
    "WRO": 3.0, # Run Out
    "WMRO": -1.0, # Missed Run Out
    "WDH": 1.5, # Direct Hit
}
```

```
# -----
# 2. Raw ball-by-ball fielding data (Dataset Features from the brief)
# pick in {"Clean Pick", "Good Throw", "Catch", "Drop Catch", "-"}
# throw in {"Run Out", "Missed Run Out", "Stumping", "Missed Stumping", "-"}
# direct_hit in {"Y", "N"}
# -----
```

```
MATCH_NO = "T20-014"
INNINGS = 1
VENUE = "Wankhede Stadium, Mumbai"
FIELDING_TEAM = "Falcons XI"
```

```
RAW_EVENTS = [
    # ballcount, position, short description, pick, throw, direct_
    hit, runs, over, player
    (1, "Cover", "Fielded cleanly, quick return", "Clean Pick", "-", "N", 0, 1, "R
    ahul Sharma"),
    (2, "Mid-off", "Dived to save a certain boundary", "Clean Pick", "-", "N", 2,
    1, "Steve Walker"),
    (3, "Point", "Sharp low catch taken", "Catch", "-", "N", 0, 1, "Liam
```

O'Brien"),

(4, "Backward Pt", "Direct hit runs out the batter", "Clean Pick", "Run Out", "Y", 1, 2, "Rahul Sharma"),

(5, "Long-on", "Misfield, ball raced to boundary", "-", "-", "N", -4, 2, "Steve Walker"),

(6, "Mid-wicket", "Good pick, flat throw to keeper", "Clean Pick", "-", "N", 1, 2, "Liam O'Brien"),

(7, "Cover", "Dropped a regulation catch", "Drop Catch", "-", "N", -2, 3, "Rahul Sharma"),

(8, "Point", "Excellent diving stop", "Clean Pick", "-", "N", 1, 3, "Steve Walker"),

(9, "Slip", "Sharp catch off an edge", "Catch", "-", "N", 0, 3, "Liam O'Brien"),

(10, "Mid-off", "Fumble under pressure", "-", "-", "N", -1, 4, "Rahul Sharma"),

(11, "Long-off", "Good throw, close run out chance", "Good Throw", "Missed Run Out", "N", 0, 4, "Steve Walker"),

(12, "Backward Pt", "Direct hit, missed by inches", "Clean Pick", "Missed Run Out", "Y", 0, 4, "Liam O'Brien"),

(13, "Cover", "Fielded and returned accurately", "Good Throw", "-", "N", 1, 5, "Rahul Sharma"),

(14, "Keeper", "Quick stumping off a wide delivery", "Clean Pick", "Stumping", "N", 1, 5, "Steve Walker"),

(15, "Mid-wicket", "Clean stop, no run conceded", "Clean Pick", "-", "N", 0, 5, "Liam O'Brien"),

(16, "Point", "Diving catch inches from ground", "Catch", "-", "N", 0, 6, "Rahul Sharma"),

(17, "Long-on", "Overthrow after sloppy pick", "-", "-", "N", -3, 6, "Steve Walker"),

(18, "Backward Pt", "Direct hit run out, brilliant work", "Clean Pick", "Run Out", "Y", 1, 6, "Liam O'Brien"),

(19, "Cover", "Good throw kept the batters to one", "Good Throw", "-", "N", 1, 7, "Rahul Sharma"),

(20, "Slip", "Dropped a tough low chance", "Drop Catch", "-", "N", -2, 7, "Steve Walker"),

]

```

# -----
# 3. Build the full dataset (adds Match No, Innings, Team, Venue to each row)
# -----
def build_dataset():
    rows = []
    for (ballcount, position, desc, pick, throw, direct_hit, runs, over, player) in RAW_EVENTS:
        rows.append({
            "Match No": MATCH_NO,
            "Innings": INNINGS,
            "Team": FIELDING_TEAM,
            "Player Name": player,
            "Ballcount": ballcount,
            "Position": position,
            "Short Description": desc,
            "Pick": pick,
            "Throw": throw,
            "Direct Hit": direct_hit,
            "Runs": runs,
            "Overcount": over,
            "Venue": VENUE,
        })
    return rows

```

```

# -----
# 4. Aggregate raw events into per-player metrics and compute Performance Score
# -----
def compute_performance(rows):
    metrics = defaultdict(lambda: {
        "CP": 0, "GT": 0, "C": 0, "DC": 0,
        "ST": 0, "RO": 0, "MRO": 0, "DH": 0, "RS": 0,
    })

```

```

for r in rows:
    m = metrics[r["Player Name"]]
    if r["Pick"] == "Clean Pick":
        m["CP"] += 1
    elif r["Pick"] == "Good Throw":
        m["GT"] += 1
    elif r["Pick"] == "Catch":
        m["C"] += 1
    elif r["Pick"] == "Drop Catch":
        m["DC"] += 1

    if r["Throw"] == "Run Out":
        m["RO"] += 1
    elif r["Throw"] == "Missed Run Out":
        m["MRO"] += 1
    elif r["Throw"] == "Stumping":
        m["ST"] += 1

    if r["Direct Hit"] == "Y":
        m["DH"] += 1

    m["RS"] += r["Runs"]

results = []
for player, m in metrics.items():
    ps = (
        m["CP"] * WEIGHTS["WCP"]
        + m["GT"] * WEIGHTS["WGT"]
        + m["C"] * WEIGHTS["WC"]
        + m["DC"] * WEIGHTS["WDC"]
        + m["ST"] * WEIGHTS["WST"]
        + m["RO"] * WEIGHTS["WRO"]
        + m["MRO"] * WEIGHTS["WMRO"]
        + m["DH"] * WEIGHTS["WDH"]
        + m["RS"]
    )

```

```
    results.append({"Player Name": player, "**m", "Performance Score": round(ps, 2)
})
```

```
results.sort(key=lambda x: x["Performance Score"], reverse=True)
return results
```

```
# -----
# 5. Output helpers: console tables, CSV files, bar chart
# -----
```

```
def print_table(rows, columns):
    widths = {c: max(len(c), max((len(str(r[c])) for r in rows), default=0)) for c in columns}
    header = " | ".join(c.ljust(widths[c]) for c in columns)
    print(header)
    print("-" * len(header))
    for r in rows:
        print(" | ".join(str(r[c]).ljust(widths[c]) for c in columns))
```

```
def save_csv(rows, columns, filename):
    with open(filename, "w", newline="", encoding="utf-8") as f:
        writer = csv.DictWriter(f, fieldnames=columns)
        writer.writeheader()
        writer.writerows(rows)
    print(f"Saved: {filename}")
```

```
def save_chart(results, filename):
    players = [r["Player Name"] for r in results]
    scores = [r["Performance Score"] for r in results]

    plt.figure(figsize=(7, 4.5))
    bars = plt.bar(players, scores, color=["#2e7d32" if s >= 0 else "#c62828" for s in scores])
    plt.title(f"Fielding Performance Score - {MATCH_NO} (Innings {INNINGS})")
```

```

plt.ylabel("Performance Score (PS)")
plt.axhline(0, color="black", linewidth=0.8)
for bar, score in zip(bars, scores):
    plt.text(bar.get_x() + bar.get_width() / 2, score, f"{score}",
             ha="center", va="bottom" if score >= 0 else "top")
plt.tight_layout()
plt.savefig(filename, dpi=150)
plt.close()
print(f"Saved: {filename}")

```

```

# -----
# 6. Main
# -----

```

```

def main():
    out_dir = os.path.dirname(os.path.abspath(__file__))

    raw_rows = build_dataset()
    raw_columns = ["Match No", "Innings", "Team", "Player Name", "Ballcount",
                   "Position", "Short Description", "Pick", "Throw",
                   "Direct Hit", "Runs", "Overcount", "Venue"]

    print("=" * 90)
    print("RAW FIELDING DATA (ball-by-ball)")
    print("=" * 90)
    print_table(raw_rows, raw_columns)

    results = compute_performance(raw_rows)
    summary_columns = ["Player Name", "CP", "GT", "C", "DC", "ST", "RO", "MRO", "DH", "RS", "Performance Score"]

    print("\n" + "=" * 90)
    print("PERFORMANCE METRICS MATRIX")
    print("=" * 90)
    print_table(results, summary_columns)

```



```

print("\n" + "=" * 90)
print("RANKING")
print("=" * 90)
for i, r in enumerate(results, start=1):
    print(f"{i}. {r['Player Name']}: {r['Performance Score']}")

save_csv(raw_rows, raw_columns, os.path.join(out_dir, "raw_fielding_data.csv"))

save_csv(results, summary_columns, os.path.join(out_dir, "performance_summary.csv"))
save_chart(results, os.path.join(out_dir, "performance_scores.png"))

if __name__ == "__main__":
    main()

```

```

# -----
# SAMPLE OUTPUT (captured from running this script as-is)
# -----
"""

=====
=====
RAW FIELDING DATA (ball-by-ball)
=====
=====
Match No | Innings | Team    | Player Name | Ballcount | Position | Short Description
on       | Pick    | Throw   | Direct Hit | Runs | Overcount | Venue
-----
-----
T20-014 | 1    | Falcons XI | Rahul Sharma | 1    | Cover    | Fielded cleanly, quick return
| Clean Pick | -        | N        | 0 | 1    | Wankhede Stadium, Mumbai
T20-014 | 1    | Falcons XI | Steve Walker | 2    | Mid-off   | Dived to save a certain
boundary | Clean Pick | -        | N        | 2 | 1    | Wankhede Stadium, Mumbai
T20-014 | 1    | Falcons XI | Liam O'Brien | 3    | Point    | Sharp low catch taken
| Catch    | -        | N        | 0 | 1    | Wankhede Stadium, Mumbai

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T20-014 | 1 | Falcons XI | Rahul Sharma | 4 | Backward Pt | Direct hit runs out the batter | Clean Pick | Run Out | Y | 1 | 2 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Steve Walker | 5 | Long-on | Misfield, ball raced to boundary | - | - | N | -4 | 2 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Liam O'Brien | 6 | Mid-wicket | Good pick, flat throw to keeper | Clean Pick | - | N | 1 | 2 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Rahul Sharma | 7 | Cover | Dropped a regulation catch | Drop Catch | - | N | -2 | 3 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Steve Walker | 8 | Point | Excellent diving stop | Clean Pick | - | N | 1 | 3 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Liam O'Brien | 9 | Slip | Sharp catch off an edge | Catch | - | N | 0 | 3 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Rahul Sharma | 10 | Mid-off | Fumble under pressure | - | - | N | -1 | 4 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Steve Walker | 11 | Long-off | Good throw, close run out chance | Good Throw | Missed Run Out | N | 0 | 4 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Liam O'Brien | 12 | Backward Pt | Direct hit, missed by inches | Clean Pick | Missed Run Out | Y | 0 | 4 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Rahul Sharma | 13 | Cover | Fielded and returned accurately | Good Throw | - | N | 1 | 5 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Steve Walker | 14 | Keeper | Quick stumping off a wide delivery | Clean Pick | Stumping | N | 1 | 5 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Liam O'Brien | 15 | Mid-wicket | Clean stop, no run conceded | Clean Pick | - | N | 0 | 5 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Rahul Sharma | 16 | Point | Diving catch inches from ground | Catch | - | N | 0 | 6 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Steve Walker | 17 | Long-on | Overthrow after sloppy pick | - | - | N | -3 | 6 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Liam O'Brien | 18 | Backward Pt | Direct hit run out, brilliant work | Clean Pick | Run Out | Y | 1 | 6 | Wankhede Stadium, Mumbai

T20-014 | 1 | Falcons XI | Rahul Sharma | 19 | Cover | Good throw kept the b

atters to one | Good Throw | - | N | 1 | 7 | Wankhede Stadium, Mumbai
T20-014 | 1 | Falcons XI | Steve Walker | 20 | Slip | Dropped a tough low chance | Drop Catch | - | N | -2 | 7 | Wankhede Stadium, Mumbai

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PERFORMANCE METRICS MATRIX

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Player Name | CP | GT | C | DC | ST | RO | MRO | DH | RS | Performance Score

Liam O'Brien | 4 | 0 | 2 | 0 | 0 | 1 | 1 | 2 | 2 | 15.0
Rahul Sharma | 2 | 2 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 8.5
Steve Walker | 3 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | -5 | -2.0

=====

RANKING

=====

- 1. Liam O'Brien: 15.0
- 2. Rahul Sharma: 8.5
- 3. Steve Walker: -2.0

Saved: /home/claude/raw_fielding_data.csv
Saved: /home/claude/performance_summary.csv
Saved: /home/claude/performance_scores.png

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